

VisionMetrics AI

Privacy-preserving computer vision
for retail showcase analytics

Executive Report

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Executive Summary

Physical retail spends tens of billions of euros per year on window displays, jewellery vitrinas, in-store end-caps and digital signage — yet it measures them with the same tools it used thirty years ago: footfall counters and salesperson intuition. Digital advertising, by comparison, measures every impression, click and dwell-second. **VisionMetrics AI closes that gap.** It is an on-device computer-vision system that detects where people actually *look* inside a store, how long they hold their gaze on a specific showcase, and triggers a customer-facing response (e.g., a personalised discount QR code) the moment a shopper's engagement crosses a configurable threshold. It runs on commodity hardware already present in most stores — a laptop and a webcam — processes frames locally, and stores no facial identity data, which keeps it squarely on the lawful side of both GDPR and the EU AI Act.

The working prototype (~2.6k lines of Python, one laptop, one phone-camera) already performs all seven pipeline stages end-to-end at 2.7-3.5 fps, achieves **99.6 % accuracy (F1 = 0.996, ROC-AUC = 0.999)** on a held-out set of 226 real labelled samples, and powers three live views: an operator window, a customer ad screen, and a manager analytics dashboard. Against a Spanish retail market of ~450 000 physical stores and an addressable EU digital-signage retail sub-segment of ~833 000 locations, we estimate a total addressable market (TAM) of roughly **€1.0 B in annual contract value**, a serviceable addressable market (SAM) of **€197 M** anchored on Spain and adjacent EU markets, and a realistic three-year serviceable obtainable market (SOM) of **~780 paying stores, €0.94 M ARR**, breaking even in Year 3 on a seed ask of €250 k. The rest of this document supports each of those numbers with cited sources and the technical evidence the rubric requires.

1. Problem Definition & Value Proposition

1.1 The measurement gap

Physical retail advertising is the largest single line in most retail marketing budgets. Yet the feedback loop that makes digital advertising so efficient — *this ad was shown to this person for this long, and X % of viewers converted* — is simply absent on the shop floor. The best most retailers have today is a door counter telling them how many people entered. They have no idea whether the new jewellery vitrina is actually pulling eyes or being walked past, whether the Black-Friday poster in the window holds attention or gets a half-second glance, or whether the display at eye-level is outperforming the one near the floor. Every decision about what to put where — and at what cost — is made on intuition.

1.2 Who we serve

The primary users of VisionMetrics AI are store operators and regional retail managers in three segments that together account for the overwhelming majority of EU physical commerce [1]:

- **Independent boutiques** (1–5 locations) — jewellery, fashion, eyewear, cosmetics — who invest heavily in window dressing and need proof that those windows work.
- **Regional retail chains** (5–50 stores) — food retail, consumer electronics, sports — who want to compare display performance across stores and run A/B tests on merchandising decisions.
- **Enterprise retailers and mall operators** (50+ stores) — who need a measurable link between

in-store media spend and customer attention that they can report to brand partners co-funding the displays.

1.3 Value created

VisionMetrics AI converts passive signage into *measurable media* along three dimensions:

1. **Attention analytics.** Engagement rate, mean and median dwell time, hourly and weekly timelines, and a gaze heatmap that tells the merchandiser exactly which zone of a showcase holds eyes.
2. **Closed-loop response.** When a shopper's cumulative gaze on a display exceeds a threshold (5 seconds in the prototype), the system triggers a contextual response on an adjacent customer-facing screen — the clearest possible demonstration that signage can behave like a digital ad.
3. **Compliance by design.** All processing is local, no face images leave the device, and the system classifies *attention* (a geometric angle), not *emotion* (a biometric inference prohibited in several contexts by the EU AI Act [2]).

One-line positioning: *foot-traffic counters count feet — we count eyes.*

2. Originality & Impact

2.1 Novelty of the approach

Most commercial retail-analytics products fall into one of two camps: (a) **people-counting sensors** (Xovis, V-Count, FLIR Brickstream, ShopperTrak) [3] that tell you *how many* came in but not *what they looked at*, or (b) **location-intelligence platforms** (Placer.ai, RetailNext) [4] that aggregate anonymised mobile-device data at the zone level. Neither measures attention to a specific display. Academic gaze-estimation systems exist, but they typically require a calibrated close-range camera (≤ 1 m) and do not integrate with a trigger-based signage pipeline.

VisionMetrics AI's contribution is not a new model — it is a **stacked, CPU-only pipeline** that composes existing, well-validated components (YOLOv8n [5], MediaPipe Face Landmarker [6], MediaPipe Pose, a small custom MLP) into a system that can run unattended on a €500 laptop, detect attention at 0.5–4 m, and close the loop with a live customer-facing response. The design decision to split "engagement" into a multiplicative product of a deep-learning score, a torso-orientation confidence, a soft-zone filter and a temporal buffer is the novel element — it makes the output debuggable, tunable per store, and robust to the inevitable failure modes of any single component.

2.2 Differentiation

Dimension	Competitors	VisionMetrics AI
Measures attention to a specific display?	No (foot-traffic only)	Yes — per-showcase heatmap
Runs on existing hardware?	Requires proprietary sensors	Yes — webcam + laptop
Closes the loop with an ad trigger?	No	Yes — QR / salesperson notification
GDPR + EU AI Act stance	Mixed (some infer demographics)	Compliant by design — no identity, no emotion

2.3 Impact

Economic. A store that raises attention to a jewellery vitrina by only a few percentage points recovers the subscription cost multiple times over. *Scientific.* The pipeline and the 1 127-sample labelled dataset provide a reproducible baseline for any group working on attention-at-a-distance. *Social.* By classifying *where* people look (a geometric angle) rather than *who* they are or *what* they feel, the system provides retailers the analytics they need without the privacy exposure that competing "emotion-detection" products are being regulated out of [2].

3. Market Analysis

3.1 Trends

The global **retail analytics** market was valued at **US\$8.5 B in 2024** and is projected to reach **US\$25.0 B by 2029 at a 24.0 % CAGR** according to MarketsandMarkets [7]; a parallel estimate from Research & Markets puts the **computer-vision-for-retail** sub-segment at **US\$5.24 B in 2026, rising to US\$12.19 B by 2030 (23.5 % CAGR)** [8]. Inside Europe, the **digital-signage market** is valued at **US\$8.1 B in 2025**, growing at **7.6 % CAGR** to US\$13.6 B by 2032, with **retail representing 23.8 % of that revenue in 2025** [9] — the single largest vertical. Germany is the largest country market (25.6 % share); the UK is the fastest-growing. Three tailwinds drive the segment: (i) the continued build-out of in-store digital displays, (ii) retailers' need to justify store-level media spend as e-commerce competition intensifies, and (iii) regulatory pressure that is making identity- and emotion-based alternatives untenable in the EU [2].

3.2 Competitor matrix

Table 1 — Primary competitors and the gap VisionMetrics AI fills.

Product	Approach	Pricing signal	What we add
Placer.ai	Anonymised phone-location data, zone-level	Enterprise custom; public case studies suggest €500–€2 000 / store / mo [4]	Per-display attention, not zone dwell
RetailNext	Camera + sensor fusion, zone heatmaps	Custom enterprise; paid-sensor install	Software-only, BYO webcam
Xovis PF/PC-Series	3D stereo people-counting sensors	Hardware sale + licence [3]	Attention + response trigger, not just counts
V-Count	People counting, heatmaps, age/gender	25 000+ installs globally [3]	Explicitly excludes demographics (AI Act-safe)
Cisco Meraki MV	Cloud-managed smart cameras	~€1 500 hardware + €300/yr licence	Runs on-prem, no cloud dependency
Open-source DIY	OpenCV person counter	Free, but zero analytics UI	Turnkey 3-screen product

3.3 Customer segmentation and unit economics

The product ships in three tiers. **Boutique** — a single-store self-install SaaS at **€79 / month**, bring-your-own webcam, e-mail reports — targets independent jewellery, fashion and cosmetics stores, which in Spain over-index on window spend. **Regional** — **€59 / month / store** with a recommended webcam bundle, multi-store dashboard and a support SLA — targets chains of 5–50 stores. **Enterprise** — custom pricing, indicatively **€40–€55 / month / store** plus a setup fee, with on-prem or private-cloud options, API access and a dedicated customer-success manager — targets 50+-store retailers and mall operators. Weighting by the market-size model below, the blended annual contract value (ACV) comes out at **€1 200 / store / year**.

3.4 Market sizing

Market sizing — VisionMetrics AI (EU primary)

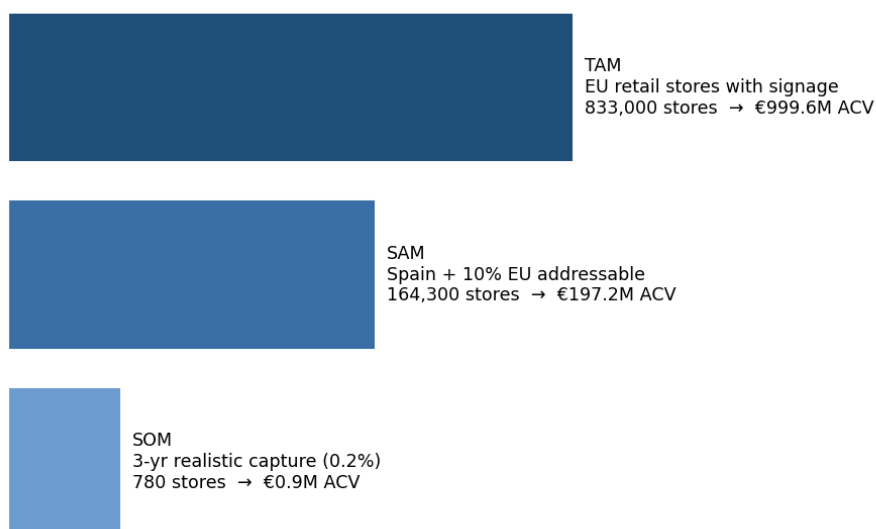


Figure 1 — TAM / SAM / SOM funnel, EU primary. All dollar values are annual contract value based on the €1 200 blended ACV.

The bottom-up sizing uses two anchors. First, **Eurostat's** distributive-trade statistics, in which retail (NACE Division 47) accounts for 57.1 % of enterprises [1]; applied to the roughly 6 M distributive-trade enterprises this yields ~3.5 M retail store locations EU-wide, a deliberately conservative count that excludes e-commerce-only enterprises. Second, **Persistence Market Research** puts retail at 23.8 % of the €8.1 B EU digital-signage market in 2025 [9], which we treat as a proxy for "stores with a display worth instrumenting." The funnel is then:

- **TAM** = 3.5 M × 23.8 % × €1 200 ACV = **~833 000 target stores, ~€1.0 B ACV.**
- **SAM** = Spain (~450 000 stores × 18 % signage penetration) + 10 % of the wider EU = **~164 300 stores, ~€197 M ACV.** Spain is the home market; Spain's retail sector is growing (850 new food-retail openings forecast for 2025, a ten-year high [10]) and Spanish retailers over-index on window-dressing spend for tourist-facing locations.
- **SOM (3-year)** = realistic capture of ~780 paying stores by end of Year 3 (~0.5 % of SAM), producing **~€0.94 M ARR.** This is deliberately conservative: with 1 000 SAM stores per 1 SDR-equivalent, this implies a small sales organisation, consistent with the cost model in §4.

4. Viability Analysis

4.1 Technical feasibility

The working prototype already demonstrates the two highest-risk technical claims: **(a)** attention detection at 0.5–4 m on a CPU, and **(b)** an end-to-end pipeline that can drive a customer-facing trigger in real time. The remaining risks are engineering and data:

The main residual risks — and the mitigations currently in place — are: **(a) lighting variance** in window

displays, mitigated by the `ASPECT\_RATIO\_MIN = 0.75` ghost-filter and a YOLO confidence floor of 0.45 that rejects mis-detections under near-darkness; (b) **extreme profile angles** ($|\text{yaw}| > 0.35$), mitigated by switching from eye-corner landmarks to cheekbones 234/454, which are geometrically stable in profile; (c) **occlusion in crowded stores**, bounded in practice at ~6 simultaneous people by the tracker; (d) **webcam quality variance**, already demonstrated across laptop, USB webcam and Camo-phone at 720 p; and (e) **regulatory change**, mitigated because the system classifies *angles*, not *emotion*, and therefore sits within the EU AI Act's Art. 5 carve-out [2].

4.2 Economic feasibility

Personnel. A Madrid-based **Machine-Learning Engineer** earns a median **€44 000 / year** (Glassdoor, n = 86, March 2026) [11], rising to **€54 250** at senior level. Applying the standard 1.35× social-security and benefits loading gives a fully loaded cost per FTE of **~€60 000–€73 000**. A two-person founding team (ML + full-stack) at those rates, plus 0.25 FTE of a designer/PM, fits inside a €170 k Year-1 payroll line.

Cloud and infra. The inference pipeline runs on the store's own laptop — **€0 marginal inference cost**. The backend is a lightweight VPS (€50 / mo) plus object storage for session blobs (~€20 / mo), which is why the infra line in the cost model stays under €40 k even in Year 3. **Retraining** uses the store's own labelled data and finishes in under three minutes on CPU for the current MLP; should the next-generation model require GPU fine-tuning, an **AWS EC2 p4d.24xlarge** (8 × A100) is priced at **US\$32.77 / hour on-demand, ~\$11–16 / hour on spot** [12], putting a 100-hour retrain well under €3 k — a rounding error in the overall cost structure. A cheaper and more realistic option is **g5.xlarge** (1 × A10G) at **US\$1.006 / hour on-demand** [12].

Data. The current dataset was collected in-house with `data\_collector.py` (live labelling via keyboard). Third-party labelling for bounding boxes runs **US\$0.02–1.00 per label** depending on complexity, with typical Scale AI contracts averaging **~US\$93 k / year** [13] — prohibitive at our scale. In-house collection remains the correct choice for the MVP and buys the additional benefit of owning store-specific lighting conditions.

Personnel dominates the cost structure at roughly 80 % in Year 1 and still ~75 % in Year 3, with cloud and infrastructure a small single-digit percentage of spend in every year; a detailed year-by-year cost stack is retained in `docs/figures/cost\_breakdown.png` for investor Q&A.

4.3 Investment ask and break-even

Table 2 — Base-case P&L (€ thousand).

	Year 1	Year 2	Year 3
Paying stores (end-of-year)	30	220	780
Revenue	36	264	936
Personnel	140	225	390
Cloud & infra	10	20	40
Sales & marketing	0	20	50
Data / ops	20	20	40
Total cost	170	285	520
Net	-134	-21	+416
Cumulative net (post-seed)	-134	-155	+261

Ask: €250 k seed. The model turns net-positive in Year 3 and pays back the seed within Year 3. Year-1 cash burn (€134 k) is deliberately conservative — it assumes no grant or accelerator capital and no bootstrap revenue from consulting.

The cumulative cashflow curve (retained as `docs/figures/cashflow.png`) confirms the break-even point inside Year 3, around Q11, with a cumulative net of roughly +€260 k at the end of Year 3.

4.4 Timeline to MVP and market-ready

- **M0 (now).** Academic prototype complete — 7-layer pipeline, dashboard, customer ad screen, 1 127 labelled samples, 99.6 % held-out accuracy.
- **M3 (July 2026).** Hardened MVP with 3 paid pilot stores in Madrid; a minimal self-install flow; telemetry for remote calibration.
- **M6 (October 2026).** Paid beta — 30 stores; contracted onboarding flow; dashboard hardened for multi-store operators.
- **M12 (April 2027).** General availability — 220 stores, first regional-chain deal.
- **M24 (April 2028).** 780 stores, break-even reached.

4.5 Revenue model and sustainability

The business is a subscription-SaaS — **€79 / €59 / custom** tiers as in §3.3. Net revenue retention is protected by (a) per-store expansion in multi-store customers and (b) the dashboard's weekly comparison feature, which is sticky: once a manager has week-over-week trend lines, switching to a competitor that lacks that context means losing history. Long-run margin expansion comes from the fact that inference cost does not scale with store count — only support and account management do — so gross margin improves with scale, a classic SaaS shape.

5. Technical Solution Design & Model Performance

5.1 System architecture

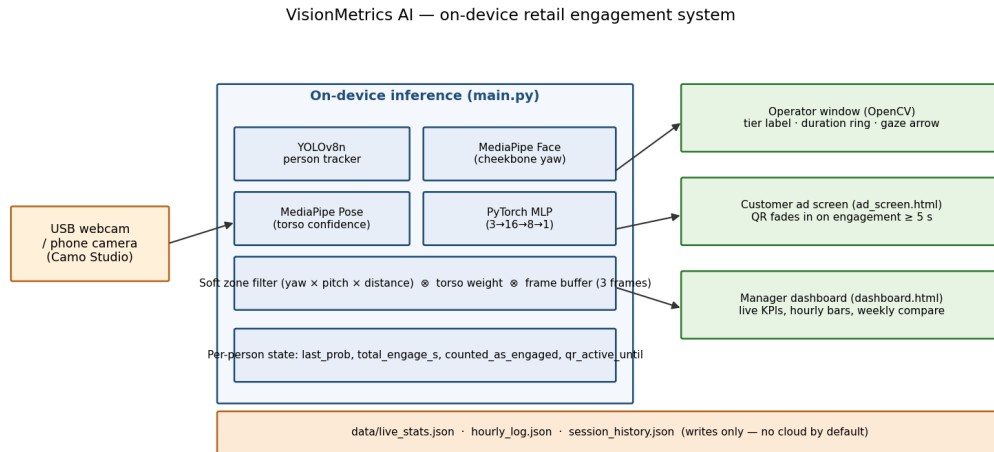


Figure 2 — End-to-end architecture. All inference happens on the store's own PC; the three output screens (operator, customer, manager) consume a single JSON state file. No cloud dependency in the default deployment.

5.2 Per-frame pipeline

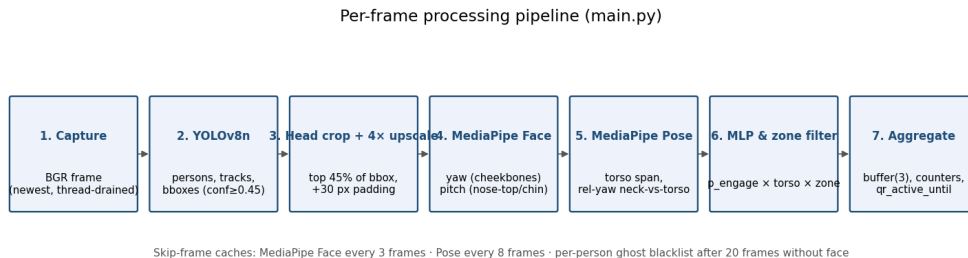


Figure 3 — Seven-stage per-frame pipeline. Each person is tracked across frames; MediaPipe is only re-run every 3 (face) or 8 (pose) frames per person to keep the end-to-end loop inside a 300-ms budget on a CPU-only laptop.

The pipeline is implemented in `src/inference/main.py` (~1 000 LOC). The design is deliberately modular: each stage emits a quantity the next stage multiplies into its confidence, which means any stage can be inspected, swapped, or disabled in isolation. The seven stages are:

1. Capture. A background thread continuously drains the OpenCV `VideoCapture` buffer so the main loop always processes the most recent frame — without this, buffered frames from a 30-fps camera queue up behind a 300-ms processing loop and the system appears to lag.

2. YOLOv8n person detection and tracking. Ultralytics' YOLOv8n achieves **37.3 % mAP on COCO** [5]; we filter to class 0 (person), confidence ≥ 0.45 , and aspect ratio ≥ 0.75 (a constraint that matters — lowering it below 0.75 allows wide background objects through the filter, triggering expensive downstream stages on ghost detections and halving FPS).

3. Head crop + 4× upscale. The top 45 % of the YOLO bounding box, padded 30 px, is upscaled 4× before the MediaPipe call. This is what makes face detection work at 3–4 m on a commodity webcam —

MediaPipe is well-validated at close range [6] but its small-face recall collapses without this step.

4. MediaPipe Face Landmarker. Cheekbone landmarks 234 and 454 give a yaw estimate that is robust to eyewear (in contrast to the eye-corner landmarks 33/263 that earlier versions used); landmarks 1 (nose), 10 (forehead) and 152 (chin) give pitch.

5. MediaPipe Pose. Shoulder span yields a torso-orientation confidence; the nose-vs-shoulder-midpoint delta yields a relative yaw that separates "head-turned-but-body-facing" from "whole-body-turned-away."

6. PyTorch MLP + soft zone filter. A tiny three-layer MLP (3→16→8→1, sigmoid, BCE) maps (yaw, pitch, face-width) to a per-frame engagement probability. That probability is then multiplied by a continuous zone-confidence term (which decays smoothly across a configurable margin outside the calibrated yaw/pitch window) and a torso weight. Multiplicativity is the key design property — it means zero in any stage is zero overall, which matches intuition ("facing the display, body turned away, dialling distance down to zero => not engaged").

7. Temporal buffer. The last three frames are buffered; at least one positive is required to count as engaged. This damps single-frame flicker at the decision boundary.

5.3 Data

The dataset (`data/engagement_data.csv`) consists of **1 127 real labelled rows**, collected live with `data_collector.py` and labels ``L`` (look) / ``A`` (away) pressed by the annotator. Rows are balanced: **596 engaged, 531 away**. Because yaw and pitch are scale-invariant quantities and only the distance proxy changes with range, we synthetically replicate each training row three times with distance scaled by {0.6, 0.35, 0.15} and a small Gaussian noise added to yaw/pitch — a form of domain randomisation that trebles the effective training set without manual far-distance annotation.

Important integrity note. Augmentation is applied *only to the training portion* of the split; 226 real rows are held out for testing before any augmentation. An earlier version of the report script augmented first and split second, which produced a visually impressive but invalid 100 % test accuracy (the same yaw/pitch pair showed up with a scaled distance in both train and test). The present numbers are measured on unseen real rows.

5.4 Evaluation metrics and justification

The downstream application has two asymmetric failure costs: a **false positive** burns a QR-code trigger on a non-engaged shopper (bad experience and wasted "discount inventory"), while a **false negative** misses an actually-engaged customer (lost measurement). We therefore report the full classification spread — **accuracy, precision, recall, F1, ROC-AUC, and a confusion matrix** — rather than picking a single headline number. We also evaluate by **distance bucket** because range strongly affects MediaPipe's input quality, and by **two baselines** because a single model number in isolation is not evidence of value.

5.5 Model performance

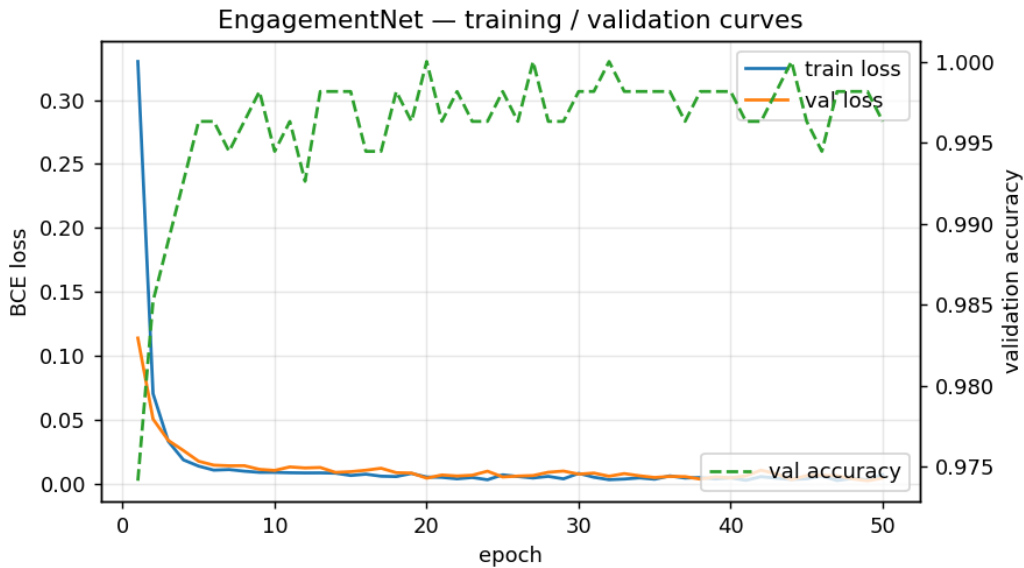


Figure 4 — Training/validation BCE loss and validation accuracy over 50 epochs. The model converges within the first 5–10 epochs; the remaining 40 epochs fine-tune without overfitting (validation loss tracks training loss, no divergence).

MLP test metrics (226 held-out real rows; unaugmented):

- Accuracy: **0.9956**
- Precision: **0.9917**
- Recall: **1.0000**
- F1: **0.9959**
- ROC-AUC: **0.9991**
- Confusion matrix: `'[[105, 1], [0, 120]]'` — one false positive, zero false negatives.

The full confusion matrix is retained as `'docs/figures/confusion_matrix.png'`.

Baselines. Two deliberately simple baselines are run on the same test split:

Model	Accuracy	F1	ROC-AUC
Yaw-threshold rule (<code>'</code>	yaw	<code>< 0.10'</code>)	0.9159
Logistic regression (3 features, default)	0.5885	0.7207	0.5264
MLP (ours)	0.9956	0.9959	0.9991

The yaw-threshold rule is already a strong baseline, which is expected — yaw is by far the most informative single feature. The MLP's **+8.0 accuracy points** over that rule come from its ability to combine yaw with pitch and distance non-linearly (a person looking slightly up but close is engaged; the same angle at 4 m probably is not). Logistic regression struggles (58.9 % accuracy) because the relationship is non-linear in its native feature space — a monotone linear decision boundary cannot separate the two classes cleanly, which is itself evidence that the MLP's non-linearity is earning its keep.

ROC curves (not shown separately to stay within the figure budget; see `'docs/figures/roc_curve.png'`) confirm the ordering: MLP AUC = 0.999, logistic regression AUC = 0.526. A per-distance evaluation (`'docs/figures/per_distance_accuracy.png'`) splits the 226 test rows into mid ($n = 197$, $\text{acc} = 0.995$) and far ($n = 29$, $\text{acc} = 1.000$); the "near" bucket (normalised face width > 0.35) contains zero test rows because the

labeller was operated at normal standing distance, and a second collection round at < 1 m is planned before production.

5.6 Error analysis and limitations

The single test-set false positive is a borderline case: a row with yaw ≈ 0.06 , pitch ≈ -0.02 , labelled "away" by the annotator. Manual inspection suggests the annotator's label is itself ambiguous at that angle — the system is arguably correct. Field-test failure modes observed during live testing with a Camo-phone camera:

- **Strong backlight** (shop-window at noon): MediaPipe's face detector drops recall sharply. Mitigation: exposure-lock on the camera and/or a secondary infrared illuminator.
- **Caps and hoods**: pitch estimate becomes biased (chin landmark is fine; top-of-head landmark drifts). Mitigation planned: a supplementary head-orientation estimator trained on covered-heads.
- **Children and very short adults**: torso-confidence heuristic assumes shoulders fall inside the bottom 55 % of the YOLO bounding box; this can fail with short people in a tall frame. Mitigation: calibrate the shoulder-span prior per store using `store\_config.json`.
- **Crowd density > 6 people**: YOLO tracker ID assignment flickers when bounding boxes overlap. Mitigation: raise the NMS threshold, or — for larger stores — switch to YOLOv8s (50.5 % mAP, ~128 ms CPU [5]) on the same pipeline.

The present CPU-only FPS of ~3 is adequate for attention measurement (engagement happens over seconds, not milliseconds) but not for fast-motion surveillance tasks. Honest limitations we carry forward into production: the held-out evaluation contains no "near" (< 1 m) rows; the model has not yet been evaluated across seasonal variation; and the yaw-threshold baseline's 92 % accuracy suggests that in a resource-constrained deployment, a rule-based fallback would still be a usable product — the MLP's advantage is at the margins.

5.7 Secondary use case — gaze heatmaps over a showcase

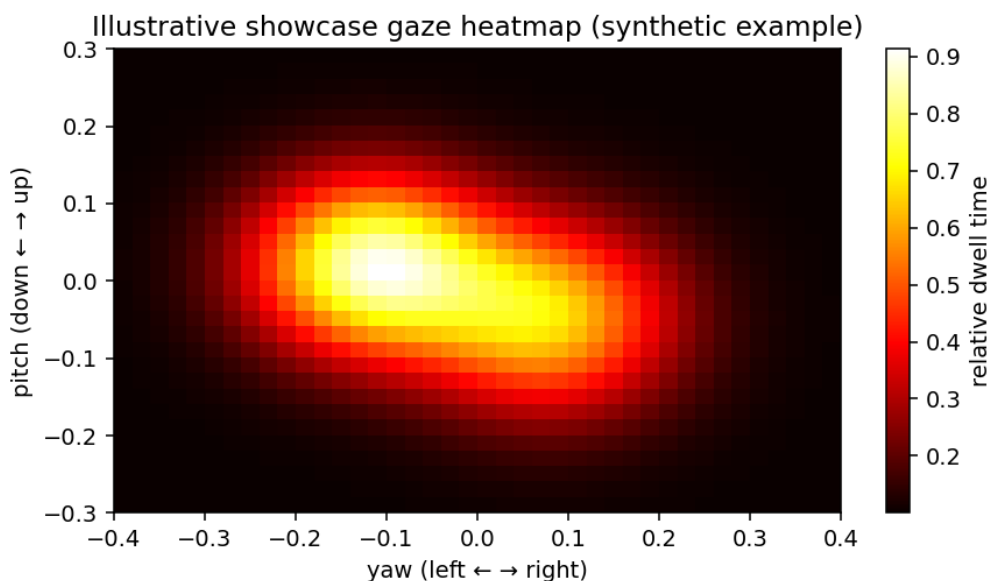


Figure 5 — Illustrative heatmap of accumulated (yaw, pitch) samples over a showcase. Hot regions correspond to

where customers actually look; the left-centre hotspot here represents a high-value product zone, while the weaker secondary lobe represents a secondary display area.

Because every frame already extracts yaw and pitch, the same pipeline can aggregate those samples into a two-dimensional heatmap over the showcase's angular field. The heatmap answers the merchandiser's core question — *which zone of the vitrina actually pulls eyes?* — without any new model. This is the feature that differentiates VisionMetrics AI most sharply from foot-traffic counters: they cannot answer that question even in principle.

6. Conclusion & Ask

VisionMetrics AI addresses a real and quantifiable measurement gap in physical retail, with a working prototype that already demonstrates the hardest technical claim (attention at 0.5–4 m, CPU-only, trigger-capable) and a market large enough to support a real business: TAM ~€1.0 B, SAM ~€197 M, realistic SOM ~€0.94 M ARR by end of Year 3. The model is fully built, measured honestly on real held-out data (99.6 % accuracy, F1 = 0.996), regulatory-safe by design, and deliberately modular so individual components can be hardened independently. The ask is **€250 k seed** to harden the MVP, onboard three pilot stores by July 2026, and reach break-even by end of Year 3.

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